AlphaPaaS Availability & Resilience Standard

Reference: ARS · Version 3.2 · Effective 1 January 2026

Document owner	AlphaPaaS Cloud Broker — Architecture Governance
Approved by	AlphaInsure Group Architecture Board
Review cycle	Annual
Previous version	3.1 (January 2025)
Classification	Internal
Related standards	AHS, CKS, FIN, ASC, DCH

1. Purpose and scope

This standard defines the availability and resilience requirements for applications hosted on the AlphaPaaS container platform. It is the authoritative reference for architecture reviews, production-readiness assessments, and incident post-mortems.

The standard applies to all workloads deployed in production and pre-production application zones on AlphaPaaS, regardless of the underlying cloud service provider (AWS or Azure). It does not apply to developer sandboxes or to workloads running outside AlphaPaaS.

Compliance with this standard is mandatory for Silver and Gold tier workloads. Bronze tier workloads should treat the Silver requirements as recommended practice.

2. Definitions

For the purpose of this document:

- **Criticality tier** — the Bronze, Silver, or Gold classification assigned to an application based on business impact. The tier dictates the availability and resilience controls that apply.
- **Availability Zone (AZ)** — a physically isolated location within an AlphaPaaS platform region. AlphaPaaS platforms currently operate across three AZs per region.
- **Recovery Time Objective (RTO)** — the maximum acceptable time between the onset of an incident and the restoration of service.
- **Recovery Point Objective (RPO)** — the maximum acceptable period of data loss, measured from the point of the incident backwards.
- **Pod Disruption Budget (PDB)** — a Kubernetes resource that limits the number of concurrent voluntary disruptions a workload can tolerate.

3. Criticality tiers

Every application hosted on AlphaPaaS must be assigned a criticality tier. The tier is declared in the application's intake form and validated by the Cloud Broker during architecture review. Tier requirements are cumulative: Gold workloads must meet all Silver requirements in addition to Gold-specific requirements.

Requirement	Bronze	Silver	Gold
SLA (monthly)	99.0%	99.5%	99.9%
Minimum replicas	1	3	5
Multi-AZ distribution	Optional	Required	Required

Requirement	Bronze	Silver	Gold
Multi-region DR	Not required	Not required	Required
PodDisruptionBudget	Not required	Required	Required
HorizontalPodAutoscaler	Optional	Required	Required
Topology spread across zones	Optional	Required	Required
Readiness probe	Required	Required	Required
Liveness probe	Recommended	Required	Required
Startup probe	Optional	Recommended	Required
Recovery Time Objective	Best effort	4 hours	1 hour
Recovery Point Objective	24 hours	1 hour	15 minutes
Database backup retention	7 days	30 days	90 days
DR exercise cadence	Not required	Annual	Biannual
Production change window	Any time	Business hours	Approved windows only

Note: Silver tier requirements are highlighted because Silver is the default tier for customer-facing internal applications.

INFO — Tier assignment

Tier assignment is a business decision, not a technical one. It reflects the consequences of unavailability to the business, not the complexity of the workload. Applications that process financial transactions, regulatory submissions, or customer-facing quotations are expected to be Silver or Gold.

4. Deployment requirements

This section defines the baseline deployment topology for workloads at each tier. Requirements are expressed as clauses that may be referenced in architecture reviews and audit findings.

ARS-01 Every application deployed to AlphaPaaS MUST declare its criticality tier in the application intake form before first production deployment.

ARS-02 Production deployments MUST meet all requirements of their declared tier. Deployments that do not meet tier requirements MUST NOT be promoted to production without an approved exception (see Section 13).

ARS-03 Silver and Gold tier workloads MUST be deployed using the `Deployment` or `StatefulSet` resource. The legacy `DeploymentConfig` resource MUST NOT be used for new deployments, and existing usage MUST be migrated at the next major release of the application.

ARS-04 Silver and Gold tier workloads MUST define a `PodDisruptionBudget` in the same namespace as the workload. The `minAvailable` value MUST be set such that the workload retains capacity for its baseline load during voluntary disruptions: `minAvailable = replicas − 1` for Silver, `replicas − 2` for Gold.

ARS-05 Silver and Gold tier workloads MUST distribute pods across availability zones using `topologySpreadConstraints` with `topologyKey: topology.kubernetes.io/zone` and `maxSkew: 1`. Node anti-affinity rules alone do not satisfy this requirement.

5. Health probes

Health probes allow the AlphaPaaS platform to route traffic correctly, detect stuck pods, and perform safe rolling updates. Probes must reflect actual application health, not merely process liveness.

ARS-06 All workloads MUST define a readiness probe that returns success only when the pod is able to serve traffic. The probe MUST exercise at least one downstream dependency critical to request handling (for example, database connectivity).

ARS-07 Silver and Gold tier workloads MUST define a liveness probe. The liveness probe MUST be lighter than the readiness probe and MUST NOT exercise external dependencies, as this creates cascading failure risk.

ARS-08 Workloads with initialisation times exceeding 10 seconds SHOULD define a startup probe to prevent premature liveness failures during cold starts. This is mandatory for Gold tier.

IMPORTANT — Common anti-pattern

Using the same endpoint for liveness and readiness probes, or having the liveness probe check a database, commonly causes a single downstream failure to trigger a cluster-wide pod restart storm. Keep liveness probes simple and local.

6. Resource management

ARS-09	All containers MUST declare both `resources.requests` and `resources.limits` for CPU and memory. Workloads without limits will be rejected by the AlphaPaaS admission controller at deployment time.
ARS-10	Gold tier workloads MUST configure memory requests equal to memory limits, to ensure Guaranteed Quality-of-Service class scheduling.
ARS-11	CPU limits above 4000m per container require a waiver from the Cloud Broker and must be justified by benchmark evidence.

7. Autoscaling

Silver and Gold tier workloads must be able to absorb traffic variation without manual intervention. This is achieved through a combination of horizontal pod autoscaling and properly sized reservations.

ARS-12	Silver and Gold tier workloads MUST define a `HorizontalPodAutoscaler` targeting the workload's primary `Deployment`. The HPA MUST use resource-based metrics (CPU and/or memory) at a minimum; custom metrics are permitted in addition.
ARS-13	The HPA `minReplicas` value MUST be no lower than the tier minimum (3 for Silver, 5 for Gold). The HPA `maxReplicas` value MUST provide at least twice the documented peak throughput capacity.
ARS-14	Workloads that rely on in-memory state (including in-process rate limiters, session caches, or local counters) MUST externalise that state before scaling horizontally. Using an in-memory rate limiter with `replicas > 1` is a breach of this clause.

8. Backup

Backup of application state is an application-level responsibility. The AlphaPaaS platform does not back up Persistent Volume contents automatically. Application teams are responsible for configuring and testing their own backup mechanisms.

ARS-15	Application databases MUST have automated backups enabled with retention matching the tier: 7 days for Bronze, 30 days for Silver, 90 days for Gold. Backup encryption at rest is mandatory regardless of tier.
ARS-16	Backups MUST be stored in a location that survives the loss of the primary availability zone. For Gold tier, backups MUST additionally be replicated to a second region.
ARS-17	Silver and Gold tier workloads MUST perform a successful restore test at least once per quarter. The outcome of the test MUST be recorded in the application's run book.

9. Disaster recovery

AlphaPaaS is designed to survive the loss of a single availability zone without manual intervention, provided that workloads are correctly configured. Loss of a full region is considered a total loss: applications requiring regional resilience must be Gold tier and architected for cross-region failover.

ARS-18 Silver tier workloads MUST be able to continue operating with the loss of any single availability zone. This includes their data stores: a single-AZ database backing a Silver workload is a breach of this clause.

ARS-19 Gold tier workloads MUST be able to fail over to a secondary AlphaPaaS region within the declared RTO. The failover procedure MUST be documented and executable without access to the failed region.

NOTE — Azure managed databases

Azure Database for PostgreSQL Flexible Server deployed without zone redundancy is bound to a single availability zone. Silver tier workloads using this service MUST enable zone-redundant high availability. Single-zone deployments are permitted only for Bronze tier and for pre-production environments.

10. Disaster recovery testing

ARS-20 Silver tier workloads MUST undergo a documented DR exercise at least once every twelve months. Gold tier workloads MUST undergo a documented DR exercise at least once every six months.

ARS-21 DR exercises MUST include, at minimum: (a) simulated loss of one availability zone; (b) verification that surviving replicas absorb traffic within the declared RTO; (c) verification that data written during the exercise is retained after recovery.

ARS-22 The outcome of each DR exercise MUST be filed with the Cloud Broker within ten business days of the exercise. Failed exercises MUST be accompanied by a remediation plan.

11. Deprecated platform resources

The following AlphaPaaS and Kubernetes resources are deprecated. They MUST NOT be used for new deployments, and existing usage MUST be tracked as technical debt with a scheduled migration.

Deprecated resource	Replacement	Migration deadline
DeploymentConfig	Deployment	End of Q4 2026
BuildConfig (Source strategy)	Tekton Pipelines or external CI	End of Q2 2027
Storage class gp2.*	gp3.regular / gp3.iops-*	End of Q2 2026
Storage class io1.*	io2.iopspergb-*	End of Q2 2026
Route without explicit router label	Route with router=internal/external/irp/waf	End of Q4 2026
Secret values passed via envFrom	Secret mounted as file volume	End of Q4 2026

ARS-23 Use of any resource listed in the deprecated resources table on a Silver or Gold tier workload MUST be remediated by the stated migration deadline. The Cloud Broker will track compliance as part of the annual architecture review.

12. Exceptions and waivers

Exceptions to this standard are granted by the Cloud Broker through the standard waiver process. Waivers are time-bound (maximum twelve months) and must be accompanied by a mitigation plan. Waivers cannot be granted for clauses that derive from Group Security, Data Protection, or regulatory obligations.

Applications operating under an active waiver are flagged in the architecture review report and tracked on the Cloud Broker's technical debt register.

13. Related standards

- **AHS** — AlphaPaaS Application Hosting Standard (exposure, networking, routing)
- **CKS** — AlphaPaaS Container & Kubernetes Security Standard
- **FIN** — AlphaPaaS FinOps & Cloud Cost Standard
- **ASC** — AlphaPaaS Approved Services Catalog
- **DCH** — AlphaPaaS Data Classification & Handling Standard

End of document. Questions regarding this standard should be directed to the AlphaPaaS Cloud Broker via the internal service desk.